

MARL+Patrolling

Design of the GNN.

We must first specify the nodes and edges. First, the nodes:

1. The system contains two types of nodes, stationary nodes (patrolling locations) and moving nodes (**agents that are on edges**).
2. Each node i has a 3-dimensional feature vector $[I_i, n_i, f_i]$, where I_i is the idleness of the node, n_i is the number of agents at the current node, and f_i is a binary variable indicating whether the node is a stationary graph node or a moving node.
 - a. NOTE: when agents are at a node i , these agents are not treated as (moving) nodes of the system, and their existence is reflected by the component n_i .
 - b. NOTE: when 2 or more agents overlap and they move together on an edge, they are treated as a single node. In this case, $I_i = 0$, n_i is the number of overlapping moving agents, and f_i is set so that they are treated as moving nodes. HOWEVER, if two agents overlap but their movement directions are different, they are treated as two separate moving nodes.
 - c. If more than 2 nodes move on an edge and they do not overlap, they are treated as two different moving nodes, with $I = 0$ and $n_i = 1$.
 - d. When an agent just leaves a node, it is treated immediately as a new moving node with $I = 0$ and $n_i = 1$.

Given any configuration of the system, the set of nodes and their associated feature vectors are clear by the above definition. In this graph, the set of (moving) nodes is changing because of the moving agents.

Having defined the nodes, we next define the set of edges.

1. For a stationary node i , the set of stationary nodes that are its neighbors is part of the neighbor set. In addition, it also has in its neighbors set the set of

INCOMING moving nodes. That is, the agents (moving nodes) that move along an edge that leads to the node i .

2. For a moving node i on an edge, the two stationary nodes that the edge connects are in its neighbor set. In addition, it also has in its neighbors set the set of INCOMING moving nodes. That is, the agents (moving nodes) that move along the same edge as i and move towards i .
3. The feature that a node j pulls from its neighbor node i contains the node feature $[I_i, n_i, f_i]$ defined before and the distance w_{ij} from i to j , i.e., the neighbor feature would be $[I_i, n_i, f_i, w_{ij}]$. Note that w_{ij} is the distance from an agent to the node if the neighbor i is a moving agent.

With the graph structure and associated features, we may apply some type of GNN to process the information.

NOTE: perhaps it is not necessary to process information at moving nodes. But if not doing so, it is not clear how the feature vectors at these nodes should be updated.